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REMOTE SENSING AND BIG GEODATA
EXERCISE 3
SPECTRAL INDEXES

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Introduction

Spectral indices allow us to extract quantitative information about the state of the Earth's surface from satellite images, going far beyond what the human eye can see. They are dimensionless values calculated by mathematically combining the spectral bands of the images in order to highlight specific physical properties of the land.
In this exercise, we had to apply two spectral indices, NDVI and MNDWI, to two distinct case studies using the Google Earth Engine (GEE) platform, with the goal of demonstrating how remote sensing can be used to monitor environmental phenomena.
The first case study focuses on vegetation monitoring following a wildfire event in Oregon (USA), while the second examines how the surface area of Lake Albano (Lazio, Italy) changed over a ten year period.

A spectral index is a mathematical equation applied pixel by pixel to the different spectral bands of a satellite image. The goal is to bring out physical characteristics of the land that would not be visible by looking at individual bands directly.
The most common formula is the normalized difference (ND), defined as:

$$ND = \frac{B1 - B2}{B1 + B2} \quad -1 \leq ND \leq 1$$

This formula has an important advantage: it minimizes, as much as possible, the effects of varying illumination, such as shadows in mountainous areas or cast by clouds, since these effects tend to influence both bands proportionally. The result always falls between −1 and +1, which makes indices comparable across different images.
Dozens of spectral indices have been developed for specific applications (vegetation, water, snow, bare soil, burned areas, geology, and more).
The two used in this exercise are the NDVI and the MNDWI.

NDVI, Normalized Difference Vegetation Index

The NDVI is the most widely used spectral index for vegetation monitoring. Its physical principle is based on the optical behaviour of leaves: chlorophyll strongly absorbs red light (RED) to support photosynthesis, while the internal structure of spongy mesophyll cells strongly reflects near-infrared light (NIR). A pixel covered by healthy vegetation therefore shows high reflectance in the NIR and low reflectance in the RED; the normalized difference between the two bands amplifies this contrast:

$$NDVI = \frac{NIR - RED}{NIR + RED} \quad -1 \leq NDVI \leq 1$$

```
//NDVI (manual way)
//num
var nominator = NIR.subtract(RED)
//denom
var denom = NIR.add(RED)
var NDVI = nominator.divide(denom)

//(real way)
var NDVI_real = S2_filtered_scaled_median.normalizedDifference(['B8','B4'])
Map.addLayer(NDVI_real,{palette:['white','green'],min:-1,max:1},'NDVI_real')

//visualization
Map.addLayer(S2_filtered_scaled_median,{bands:['B8','B4','B3'],min:0,max:0.25},'S2_filtered_scaled_median8')
```

Figure 1: NDVI code in GEE

The values can be interpreted as follows:

≈ -1	Water
Between -0.1 and 0.1	Rocks, Sand or Bare Soil
Between 0.2 and 0.3	Sparse or Degraded Vegetation
> 0.3	Healthy Vegetation
≈ 1	Very Dense and Healthy Vegetation

High NDVI values correspond to healthier and denser vegetation, while low or negative values indicate the absence of vegetation or the presence of water.
On Sentinel-2, the bands used are NIR = B8 and RED = B4.

Case Study

The Holiday Farm Fire is one of the most devastating wildfires in Oregon's recent history. It burned from September 7 to October 29, 2020, across Lane and Linn counties, Oregon. The flames spread rapidly through the communities of Blue River, Finn Rock, Nimrod, Vida, and Leaburg, destroying homes, businesses, and infrastructure across a mixed conifer forest.
The fire was caused by downed power lines during an unusually strong easterly wind event, at the driest time of year in western Oregon.

Analysis and Results

The analysis was based on a collection of Sentinel-2 satellite images filtered over the area of interest at different points in time: a pre-fire acquisitions (from July 5 to August 30, 2020) and a post-fire acquisition (from September 1 to September 20, 2020). For both processing steps, a cloud cover filter was applied to exclude images with more than 50% cloud coverage, and the appropriate radiometric scaling factor was applied, taken directly from the GEE dataset, as it varies depending on the type of satellite imagery selected. The final index values were computed on the mean composite of the filtered images.

```
var S2_filtered = S2.filterDate('2020-09-01','2020-09-20')
.filterBounds(geometry)
.filter(ee.Filter.lt('CLOUDY_PIXEL_PERCENTAGE',50))

//radiometric scale
function maskS2clouds(image) {
  var qa = image.select('QA60');

  // Bits 10 and 11 are clouds and cirrus, respectively.
  var cloudBitMask = 1 << 10;
  var cirrusBitMask = 1 << 11;

  // Both flags should be set to zero, indicating clear conditions.
  var mask = qa.bitwiseAnd(cloudBitMask).eq(0)
      .and(qa.bitwiseAnd(cirrusBitMask).eq(0));

  return image.updateMask(mask)
  .divide(10000)
  .copyProperties(image,image.propertyNames());
}

var S2_filtered_scaled = S2_filtered.map(maskS2clouds)

var S2_filtered_scaled_median = S2_filtered_scaled.median()
Map.addLayer(S2_filtered_scaled_median,{bands:['B4','B3','B2'],min:0,max:0.25},'S2_filtered_scaled_median')
```

Figure 1: pre-processing code in GEE

In the pre-fire image, the entire study area is dominated by deep green tones, indicating a dense and healthy forest, with mean NDVI values across the area of interest around 0.80. The only significant interruptions correspond to watercourses (shown in white) and marginal agricultural or urbanized areas (light green), both fully consistent with the geographical context of the valley.

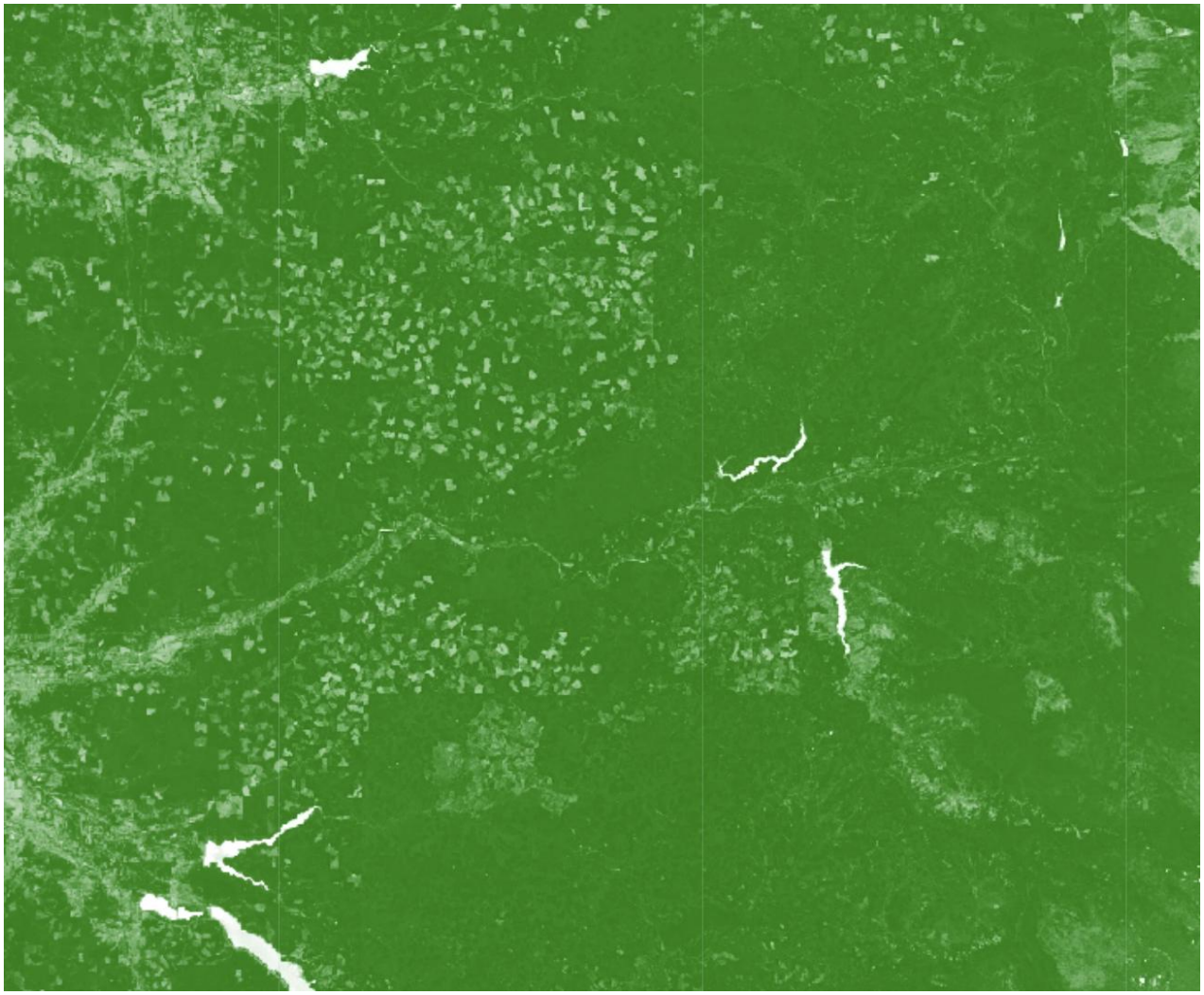


Figure 3: Pre Event NDVI

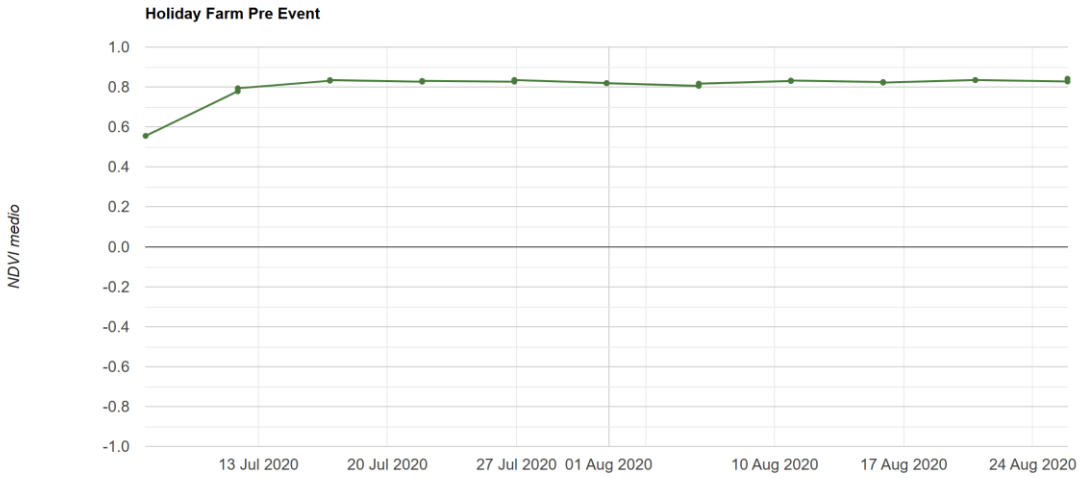


Figure 2: Pre Event NDVI chart

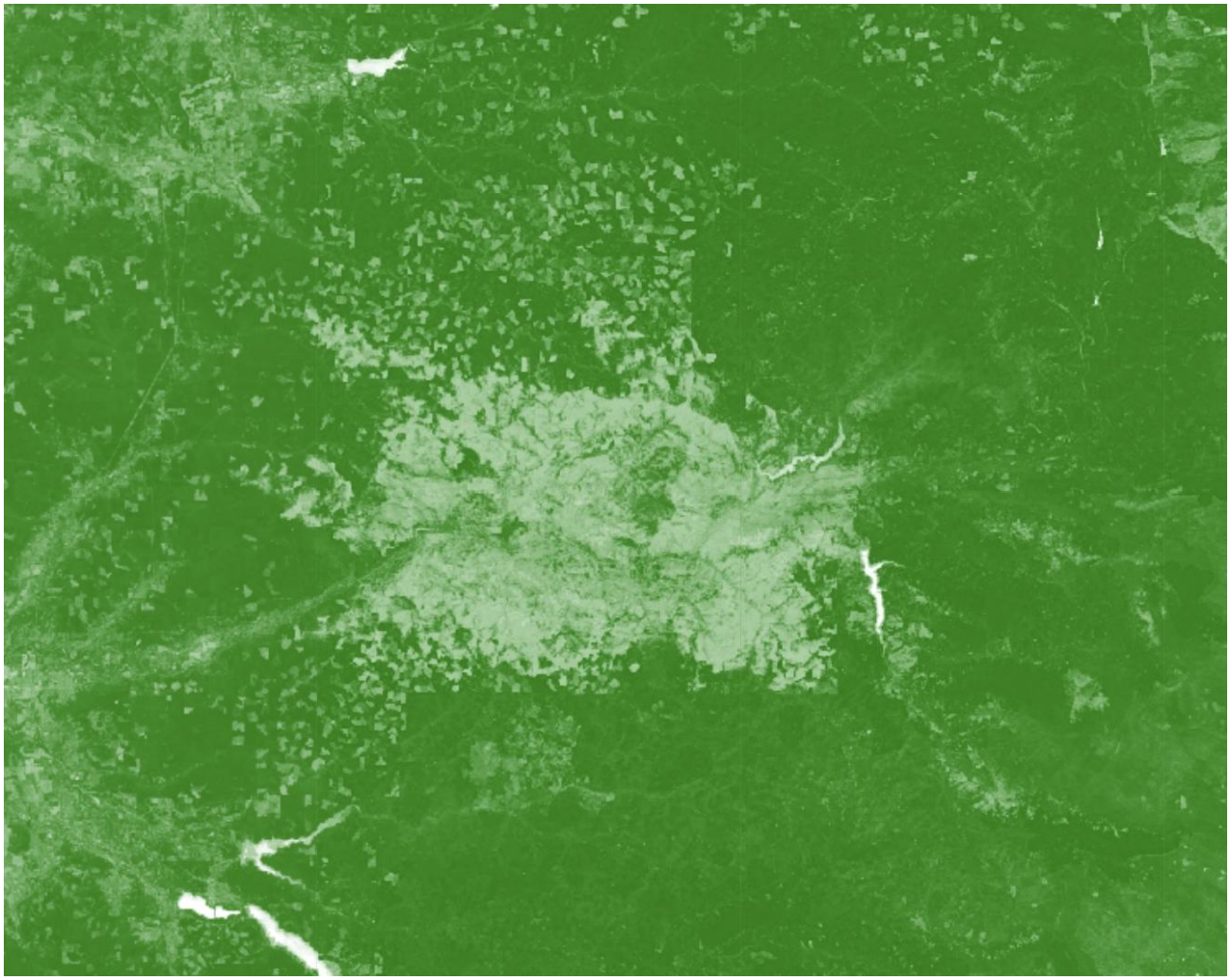


Figure 4: Post Event NDVI

In the post-fire image, the situation is radically different. The central portion of the study area, corresponding to the fire perimeter, shows a marked drop in NDVI values toward very pale green tones. The burned area stands out clearly as a large, diffuse patch of degraded vegetation, surrounded by a ring of intact forest that was not reached by the flames.

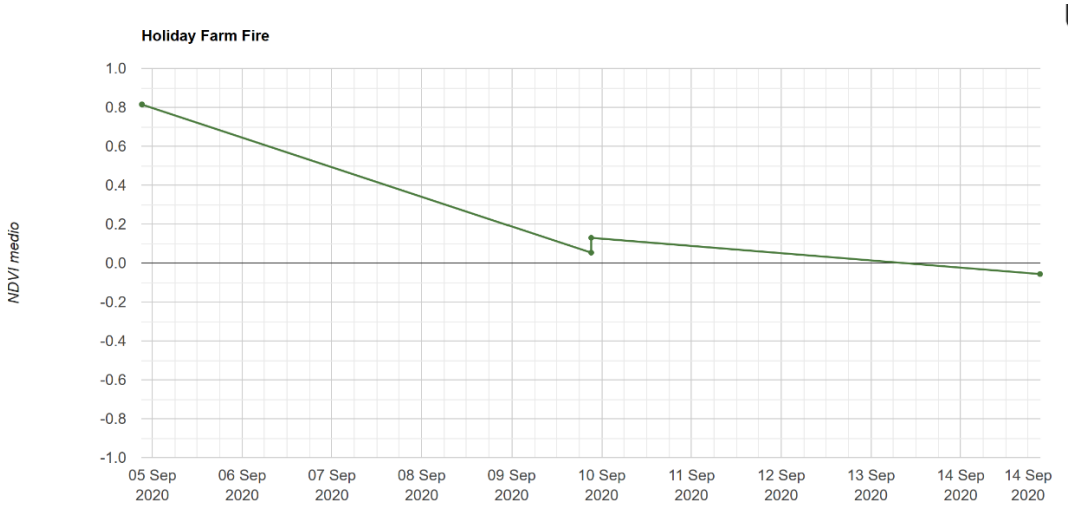


Figure 5: NDVI's drop after the fire event

As shown in the graph, on the 5 September 2020, two days before ignition, the mean NDVI over the area was approximately 0.80, consistent with what was observed earlier. From the start of the fire the curve drops rapidly and almost linearly, reflecting the progression of the fire front across the study area. Around September 10, a very brief stabilization near 0.10 is visible, likely due to reduced wind conditions that temporarily slowed the spread, before the decline resumed in the following days. By September 14, the mean NDVI had already reached negative values (around −0.10), showing that the surface is dominated by burned soil and carbonaceous residues.

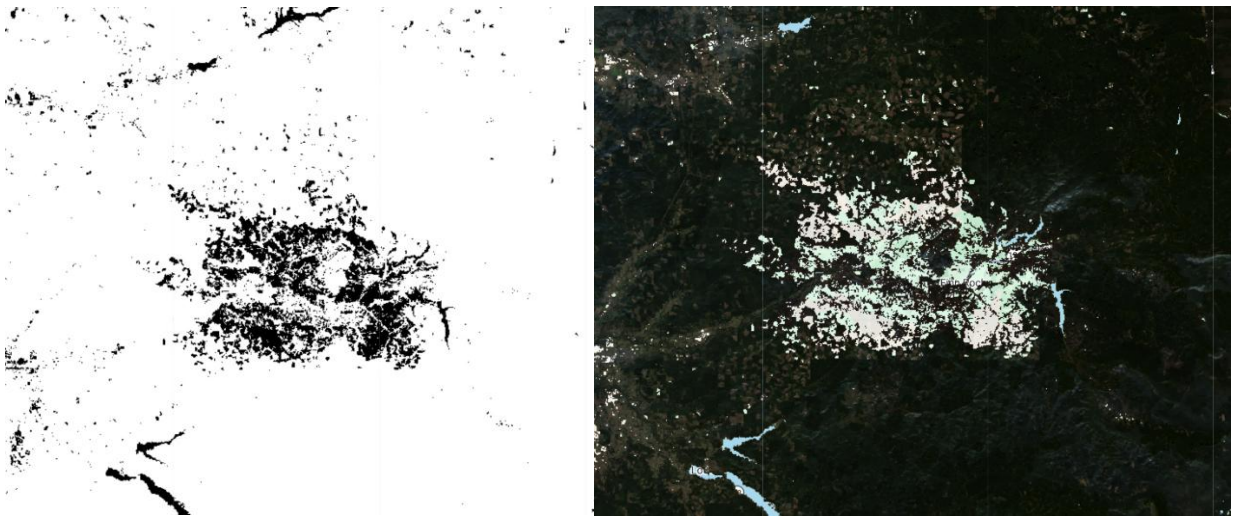


Figure 6: masked image

To further support the analysis, a quantitative estimate of healthy vegetated surface area was computed in Google Earth Engine by applying a mask based on an NDVI threshold of 0.3.

```
//NDVI > 0.3
var mask = NDVI.gt(0.3)
Map.addLayer(mask,{},'MASK')
//white pixels NDVI = 0.3, others are black

var masked_image = S2_filtered_scaled_median.updateMask(mask)
Map.addLayer(masked_image,{bands:['B4','B3','B2'],min:0,max:0.3},'masked_image')
```

Figure 7: NDVI mask code in GEE

For each of the two scenes (pre- and post-fire), the same procedure was followed: starting from the NDVI image computed on the Sentinel-2 median composite, a mask was generated using the ".gt(0.3)" function, which assigns a value of 1 to pixels with NDVI above the threshold and 0 to all others.
This mask was then applied to the original image using the ".updateMask()" command, which isolated only the pixels classified as healthy vegetation, making it possible to count them:

```
//B2 pixel number
var count = masked_image.select('B2').reduceRegion({
  //reducer
  reducer:ee.Reducer.count(),
  //geometry
  geometry:ROI,
  //scale
  scale:10,
  maxPixels:1e10,
})
print(count.get('B2'))

var area = ee.Number(count.get('B2')).multiply(10).multiply(10)
print('Healthy Area')
print(area)
```

Figure 8: Healthy area calculation in GEE

	Healthy Area (km^2)
Pre Event	803 km^2
Post Event	271 km^2

This represents a net loss of approximately 532 km², corresponding to 66% of the pre-fire vegetated area.

NDWI/MNDWI, Normalized Difference Water Index

The NDWI was developed to detect water content in vegetation and to identify water surfaces. It combines the NIR band with the short-wave infrared band (SWIR):

$$NDWI = \frac{NIR - SWIR}{NIR + SWIR} \quad -1 \leq NDWI \leq 1$$

The physical reasoning behind this is that SWIR reflectance is sensitive to water content in leaf structure and canopy, while NIR reflectance depends primarily on the internal structure of the leaf and dry matter, making it insensitive to water. Values above 0.2 generally indicate saturated soils, flooded areas, or water bodies.

An improved variant is the MNDWI, which replaces the NIR band with the green band (GREEN), achieving better separation between open water, vegetation, and urban areas:

$$MNDWI = \frac{GREEN - SWIR1}{GREEN + SWIR1} \quad -1 \leq MNDWI \leq 1$$

```
//MNDWI
var mndwi = image.normalizedDifference (['B3','B6']);

Map.addLayer (mndwi,{palette: ['FFFFFF', '000080'], min : -1 , max :1}, 'MNDWI');


Map.addLayer(image,{'bands': ['B6','B5','B3'], min:0,max:0.3},'landsat 8 false color image');

```

Figure 9: MNDWI code in GEE

The MNDWI is better suited for monitoring lakes, rivers, and open water surfaces, as it reduces the spectral confusion with built-up areas that affected the original formulation. The standard threshold for identifying water pixels is MNDWI > 0. On Landsat 8, the bands used are GREEN = B3 and SWIR1 = B6.

Case Study

Lake Albano (also known as Lake Castel Gandolfo) is a volcanic crater lake located in the "Castelli Romani" zone, approximately 25 km southeast of Rome. It is the deepest volcanic lake in Italy (around 170 m). Over recent decades, the lake has been experiencing a progressive and concerning decline in water level. The water level remained largely stable from 1950 to 1980, after which it began to fall: roughly 50 cm between 1980 and 1985, then down to −2 meters over the following decade, reaching approximately −4 meters below the original level between 1995 and 2005. In recent decades the level has dropped by around 7 meters in total, with an estimated loss of more than 40 million cubic meters of water.

The causes are multiple. The primary driver is intensive water extraction: the Sforza Cesarini well, operated by Acea Ato2, pumps water 24/7 to supply eleven municipalities in the Castelli Romani area, in a withdrawal system that never stops. This is compounded by land sealing due to extensive urbanization, which reduces soil permeability and prevents rainwater from infiltrating the aquifers that feed the lake.
The water level continues to drop by approximately 20 cm per year, and without a change of course the lake could shrink further within just a few decades.

Analysis and Results

Unlike the previous case study, this analysis was conducted on a Landsat 8 collection filtered over the period from June 1 to September 30, 2025.
The MNDWI image computed on the median composite clearly shows the distribution of water surfaces across the Castelli Romani area.

Pixels with high MNDWI values, shown in deep dark blue, unambiguously identify the two volcanic lakes present in the study area: Lake Albano, the larger water body, in the upper left of the image, and Lake Nemi, smaller and located in the lower right. The rest of the territory, covered by urban, agricultural, and forested areas, shows consistently low MNDWI values, displayed in pale blue and white.

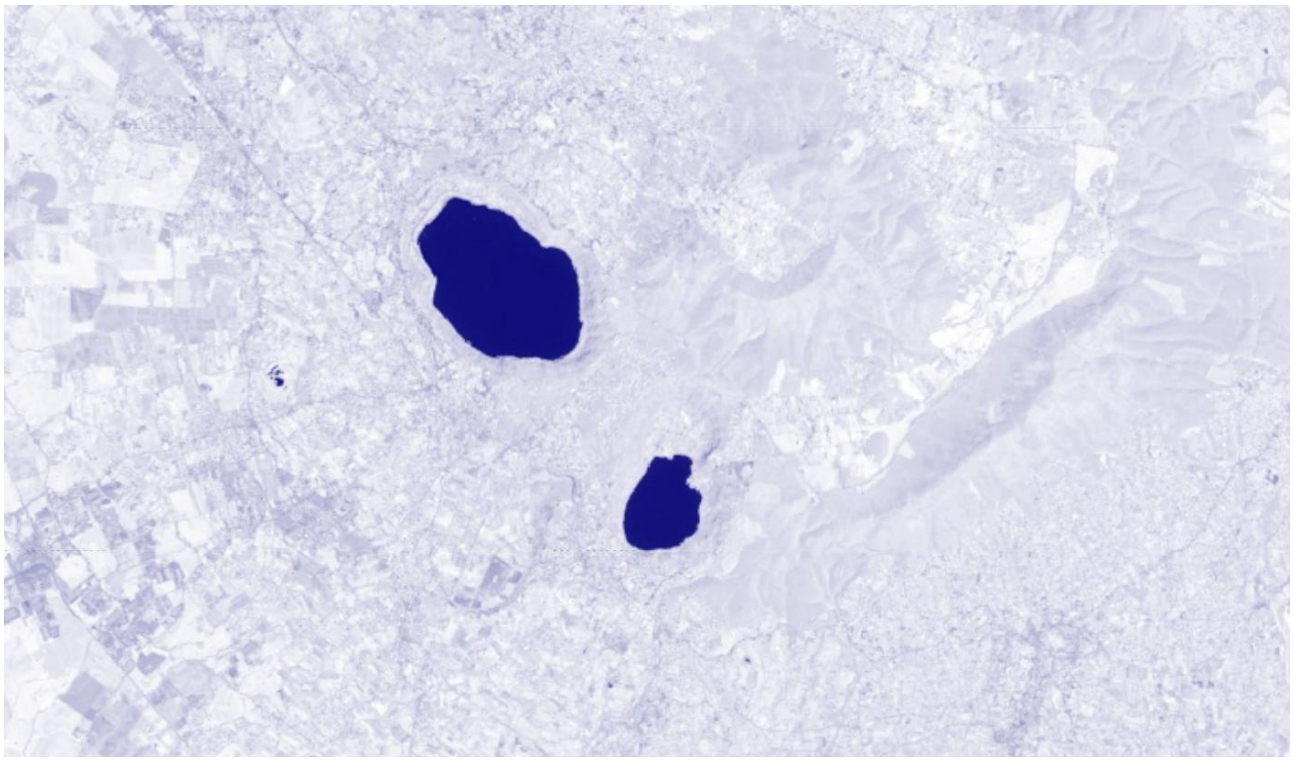


Figure 2: MNDWI layer in 2025

Focusing more specifically on Lake Albano, I've decided to carry out a ten year temporal comparison between 2015 and 2025.

```
24  //edge
25  var edge = ee.Algorithms.CannyEdgeDetector(mndwi,0.99) ;
26  Map.addLayer (edge.updateMask(edge),{palette:['red']} ,'reservoir edge');
```

Figure 10: edge code in GEE



Figure 3: 2025 (left) vs 2015 (right)

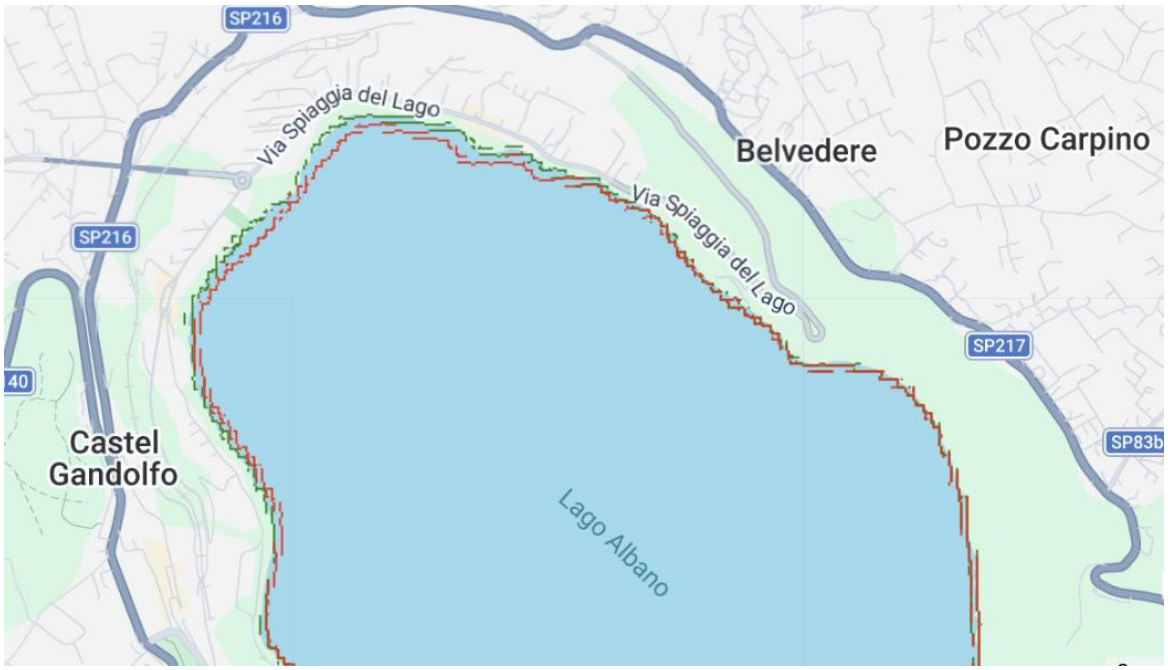


Figure 4: contour lines from 2015 (green) and 2025 (red)

The comparison between the lake outlines extracted using an MNDWI > 0 mask for the two reference years represents the most significant result of this analysis. The two perimeters are almost completely overlapping along the southern and eastern shores of the lake, where the bank is steep and rocky. The gap between the two outlines becomes visible along the northern shore, near Via Spiaggia del Lago: here the red contour (2025) is consistently set back relative to the green one (2015), indicating a reduction in the lake's surface area along this stretch. This is the morphologically flattest part of the basin, where even modest changes in water level translate into a visible retreat of the shoreline.

Conclusion

This exercise has shown how spectral indices represent highly effective tools for the quantitative monitoring of environmental change through satellite remote sensing.
Applying the NDVI to the Holiday Farm Fire case clearly highlighted the immediate effects of a wildfire on vegetation cover, revealing a sharp collapse in index values and an estimated loss of around 66% of the area previously characterized by healthy vegetation.
At the same time, using the MNDWI to analyze Lake Albano made it possible to accurately identify and compare the extent of the lake's surface at two distinct points in time, revealing a shoreline retreat in the lower-gradient areas. Although the changes observed are less dramatic than those associated with a sudden event such as a wildfire, they are consistent with the progressive decline in lake level documented by hydrometric measurements and demonstrate the effectiveness of the index in monitoring long-term hydrological processes.
It is therefore clear that spectral indices are essential tools for ecosystem monitoring, natural resource management, and the assessment of the impacts of climate change and human activities on the environment.